

ME301 Lab 2 Report

Group D: Edward Lee, Alicia Malek

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ABSTRACT

This report presents the implementation and analysis of autonomous navigation on a walking hexapod robot within a grid-based environment. The navigation utilizes three main components: localization, path planning, and mapping. Experimental trials and data collection measured navigation accuracy, execution time, and mapping coverage for both directed traversal and autonomous wandering. The trials demonstrated that the hexapod could localize its position, construct a grid map representation, and execute planned paths; however, there was significant error due to walking/turning gait uncertainty and sensing limitations.

I. INTRODUCTION

Autonomous robot navigation requires the ability to localize, plan, and map. Localization involves a robot self-identifying where it is on a map. Path planning is the ability to safely generate a path from a start-to-end position. Mapping is constructing a map of the surrounding environment and obstacles [1]. These abilities are essential for mobile robots that can be applied in healthcare and industrial settings. Specifically, in this lab, a hexapod robot will be examined to estimate its location, plan collision-free paths between arbitrary positions, and map its surroundings.

Localization consists of estimating a robot's position within a known map/environment. A robot could localize itself using odometry methods or identifying landmarks. Robot odometry involves data from shaft encoders to estimate its location from a reference position [1]. However, relying on solely odometry is not reliable due to noisy data or built up error. Using sensors to identify landmarks can assist the robot in determining its position relative to the landmark. These landmarks can include anything from a specific wall or a piece of furniture. As a robot moves, localization is crucial to correct any positional error and update its internal location. In this lab, localization is performed using discretized cells in a maze. The cells simplify the localization resolution, however introduces more variability and potential for position error/uncertainty. There are more advanced localization methods such as Kalman filtering, which compares sensor data with computational prediction based on their uncertainties to provide a more accurate position estimation [2].

Path planning determines finding a path from the robot's current location to a destination [1]. In this lab, path planning is performed on a discretized grid with a cost-based path generation. Specifically, a wavefront propagation algorithm was utilized to assign costs to each cell in a known map. There are additional path planning methods that each differ in computation efficiency such as A* which utilizes distance heuristics to find the shortest paths [3]. This distance heuristic allows the search to be guided rather than fully exploratory, so it can be faster. Path planning is applicable to several robots such as warehouse navigation, self-driving cars, or drones to safely avoid any obstacles and find the best path to the destination.

Finally, mapping involves constructing a representation of the environment from sensor observations. Mapping is crucial to robots when they are placed in an unknown environment and must build a map as it traverses the terrain. For example, precise mapping can be greatly beneficial for autonomous cars in new neighborhoods, vacuum cleaners, or search-and-rescue robots. There are various forms of mapping algorithms such as depth-first search, Eller's algorithm, and hunt-and-kill algorithm [4]. In this lab, the large grid resolutions allow for ease of computation, but introduces significant positional error and uncertainty.

When a robot is trying to construct a map while identifying its own location, simultaneous localization and mapping (SLAM) algorithms can be used. SLAM solves the "chicken-and-egg" situation of needing to identify positions in a map without having a map and having to map without knowing locations. There are various forms

of SLAM such as filter-based SLAM and graph-based SLAM. Filter-based SLAM utilizes filters such as Kalman filters to update a singular, internal robot state by weighing computational predictions and sensor readings. This approach is effective for smaller maps and computational efficiency. When a map gets too large, error can build up quickly and lead to inaccurate state estimations. Furthermore, filters such as the Kalman filter assumes linear models, so predictions can be inaccurate for nonlinear models [5]. Graph-based SLAM stores robot poses as nodes and motion information as edges. The graph based approach makes motion very scalable and general. However, as a map gets larger, the complexity of optimizing a graph and matrix makes it very computational intensive [5]. For this lab, the wavefront propagation with a depth-first approach will be utilized to simplify the computation power.

This laboratory assignment explores how simplified map representations and searching algorithms can produce effective hexapod navigation. By implementing localization, path planning, and mapping, on a robot, the experiments demonstrate both the practicality and challenges of autonomous navigation in real-world systems.

II. METHODS

The robot used is the hexapod from previous labs. The robot operates within a standardized grid-based coordinate system to ensure consistent localization and navigation. Position is represented as (i, j, k) , where i corresponds to the north-south axis and the j corresponds to the east-west axis. Positive i indicates motion towards south, and positive j indicates motion towards the east. The origin $(0, 0)$ is located at the top-left corner of the grid. Orientation is represented by a discrete heading index: 1 corresponds to North, 2 to East, 3 to South, and 4 to West. Each grid cell represents a 2×2 tile region on the floor, allowing the robot's motion to be mapped directly to discrete locations in the environment. An example of a grid can be shown below in Figure 1, where the top left cell is $(0,0)$.

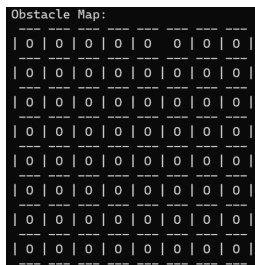


Figure 1. Example of Map Grid

To achieve localization and controlled movement, fundamental actions were implemented for walking one grid cell in each cardinal direction and turning to align with the appropriate heading. The motion control was optimized so that if the robot is already facing the desired direction, it continues moving without pausing to recompute orientation.

The system accepts user-defined start and end configurations that include position and heading. The navigation trials were from $(0, 0, 3) \Rightarrow (3, 0, 1)$, $(0, 0, 3) \Rightarrow (0, 3, 2)$, and $(0, 0, 3) \Rightarrow (2, 2, 4)$. For every motion command, the robot first traverses in the i -direction until its internal i -coordinate matches that of the command. Then, the j -direction is matched. After every step, the robot updates its internal coordinates to track its location. Each navigation task was executed 3 times, and performance was evaluated by measuring the distance error from the center of the target cell and the heading deviation from the desired final orientation. These measurements quantify the robot's navigation accuracy and reflect the cumulative effects of motion uncertainty and control precision. Execution time for each traversal was also recorded to evaluate efficiency and to assess trade-offs between speed and accuracy.

Path planning in a known environment was performed using a wavefront propagation algorithm. Planning begins at the goal cell, which is assigned a base cost of 0 and cost values are propagated outward to adjacent reachable cells with increasing values. This process continues until the start position is assigned a cost value. The robot then determines an optimal path by following a sequence of decreasing cost values from the start to the goal. As the path is traced, the corresponding grid coordinates are stored in an ordered array and reported as the best route. The generated path is converted into motion commands and executed by the robot. Three arbitrary coordinates were chosen for experimentation. Each pair of start-and-end configurations were tested 3 times. Performance was evaluated by measuring execution time and final position and heading error after traversal. These measurements assess both the effectiveness of the planning algorithm and the robot's ability to accurately execute the planned trajectory.

For mapping, the robot autonomously explored an unknown environment using a depth-first search strategy. The environment was represented as an 8x8 grid initially assumed to be fully enclosed by walls. The robot starts at (0, 0, 1). A wandering algorithm is used to make the robot walk around the maze. As the robot enters each cell, it scans all four cardinal directions by rotating its entire body towards the direction. The robot then uses a sensor to detect if a wall is in front of it if a sonar sensor returns a reading of less than 35mm. The map representation is then updated to reflect either a wall or open space in that direction. If an adjacent cell is reachable and has not yet been explored, the robot moves to that location and repeats the sensing process using a recursive function. After exploring a cell, the cell's location is marked as explored. When no unexplored paths remain down a certain path, the robot backtracks to a previous intersection and continues exploration.

Mapping performance was evaluated by recording the total execution time and the portion of the environment explored. The execution time reflects exploration efficiency and the fraction of explored cells quantifies mapping completeness. Each metric was recorded across trials to assess repeatability and consistency.

III. RESULTS

Rulers were utilized to measure the distance errors. Digital compasses and stopwatches were used for heading and time measurements respectively.

The localization tests for each pair of coordinates are recorded below in Table 1. Three trials were performed for each pair. In the x direction, north is positive and south is negative. In the y direction, east is positive and west is negative.

Table 1. Localization Tests with Distance/Heading Error and Execution Time

Goals	Distance Error (cm) (x,y)	Orientation Error (°)	Time (sec)
$x(0,0,3) \rightarrow x(3,0,1)$	(7, 0) (-1,-4) (-7,5)	-16 -22 -5	15.2 14.9 15.5
$x(0,0,3) \rightarrow x(0,3,2)$	(2.5,10) (12, 21) (10,17.5)	-18 -14 -7	12.3 12.7 12.0
$x(0,0,3) \rightarrow x(2,2,4)$	(-3.5, -3.5) (3,14) (6,20) (0,0)	-20 -2 -14	21.2 20.9 21.3

The path planning data for each pair of start and end coordinates is recorded in Table 2. Additional notes for each trial are also displayed in Table 2.

Table 2. Planning Tests with Execution Time, Distance/Heading Error, and Additional Notes

Positions $\{x_s, x_g\}$	Execution Time (sec)	Error in Final Position / Heading	Failure to Reach Goal / Why?	Notes
(3,5,1) $\rightarrow$ (1,4,1)	15	<u>Position</u> (cm): (-5, -4.5) <u>Heading</u> (degree): -3	No	Voltage: 11.0
	15	<u>Position</u> (cm): (-8, -4) <u>Heading</u> (degree): -5	No	Voltage: 10.7
(0,4,1) $\rightarrow$ (3,5,3)	19	<u>Position</u> (cm): (-8.5, -5) <u>Heading</u> (degree): -10	No	Voltage: 10.4
	18	<u>Position</u> (cm): (-7, -4) <u>Heading</u> (degree): -12.5	No	Voltage: 10.3
(1,1,1) $\rightarrow$ (2,2,3)	45 sec	<u>Position</u> (cm): (72, -11) <u>Heading</u> (degree): -12.5	Yes, 1 block short, walking distance shortened due to decrease in voltage level, didn't fully turn, tripped over plastic wall supports	Voltage: 9.9 DEMO
	47 sec	<u>Position</u> (cm): (20, 14.5) <u>Heading</u> (degree): 13	No	Voltage: 10.1 – increased walking speed compared to 1 min 40 sec trial
	1 min 40	<u>Position</u> (cm): (13, 9) <u>Heading</u> (degree): 8	No	Voltage: 10.8

The mapping data is recorded below in Table 3. In Figure 2 and 3, the initial and final mapping update is shown.

Table 3. Mapping Total Time and Portion of Map Explored

Trials	Time	Portion Explored
#1	3min, 26sec	93.75%

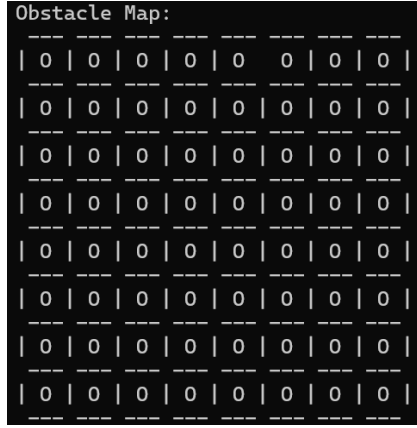


Figure 2. Initial Map

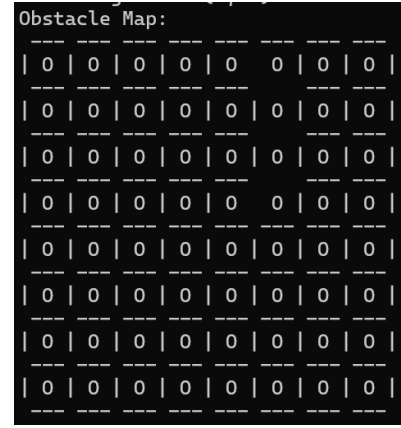


Figure 3. Final Updated Map

IV. DISCUSSION

The experimental results demonstrate that the hexapod robot was able to complete localization, planning, and mapping tasks within a structured grid environment, but performance was limited by motion uncertainty and sensing constraints. Across localization trials, consistent position and heading errors were observed after traversal to target cells. These errors were primarily attributed to gait inaccuracies associated with the tripod walking pattern. Small variations in leg placement, ground friction, and starting body orientation accumulate over successive steps, producing drift between the robot's predicted and actual position. Because localization in this system relies on discrete motion assumptions rather than continuous sensor feedback, any deviation in step length or turning angle directly propagates into position error. Furthermore, the changing battery voltage changes the power of the legs when they push to move or rotate the robot body. This can lead to further inconsistencies in the gait patterns. These variations explain the slight variability observed across the repeated trials.

Execution time measurements provide additional insight into system performance. While the optimized motion controller reduced unnecessary recomputation when the robot was already facing the desired direction, traversal speed remained constrained by the need for stable gait execution and directional alignment. Lower battery voltage also contributed to reduced walking distance and incomplete turning, as observed in planning trials where the robot failed to reach the goal or stopped short of expected motion distances. This indicates that actuation consistency is a critical factor in navigation reliability and should be considered a major source of experimental uncertainty.

Path planning using wavefront propagation successfully generated feasible routes in the known maze environment. In most trials, the robot followed the planned path and reached the intended goal region, demonstrating that the algorithm produced valid and executable trajectories. However, discrepancies between planned and executed paths highlight the difference between algorithmic optimality and physical realizability. The planning algorithm assumes ideal motion between cells, whereas the physical robot exhibits non-ideal step length and turning behavior. As a result, execution error is dominated by mechanical and control limitations rather than deficiencies in the planning method itself.

Mapping performance showed that the robot was able to explore and reconstruct a large portion of the environment using depth-first search. The exploration strategy ensured systematic coverage by prioritizing unexplored regions and backtracking when necessary. However, mapping efficiency was significantly limited by the sensing procedure. At each cell, the robot rotated its entire body to scan all four directions, which introduced substantial time overhead. A large fraction of total mapping time was spent performing orientation adjustments

rather than forward exploration. Additionally, wall contacts occurred when motion error caused the robot to approach obstacles more closely than expected, indicating limitations in distance sensing resolution and motion precision. Only 1 mapping trial was able to be conducted as the turning and the walking gait accumulated too much error to reliably provide mapping data. During the trials, sonar feedback and position correction functions were added to mitigate the error. However, the error continued to exist and had a significant impact on the walking.

Several improvements could enhance system performance. The most significant improvement would involve reducing gait-induced error through feedback control. Incorporating sensor-based correction, such as using distance measurements to the step length or heading after each move, would reduce the accumulated drift and improve localization accuracy. Other correction methods such as a fixed side step could improve overall positional accuracy as well. The hexapod has an overall rightward steady state error in the walking gait. At the end of a motion, a side step to the left could adjust the real-world position to better represent the internal position of the robot. For mapping, replacing the full body rotation with the rotating sonar sensor would dramatically reduce exploration time by allowing sensing without reorienting the robot. Improved energy management or voltage regulation would also increase motion consistency across trials.

Overall, the results illustrate that algorithmic navigation methods can function within simplified environments, but physical implementation introduces practical constraints that significantly influence performance. The experiments highlight the importance of robust motion control, reliable sensing, and efficient exploration strategies in autonomous robotic systems.

V. CONCLUSION

This laboratory implemented and evaluated localization, path planning, and mapping on a hexapod robot operating in a grid-based maze. The robot successfully executed discrete navigation actions and internal position updating, generated paths using wavefront propagation, and constructed a partial map of an unknown environment using depth-first searching. Experimental results demonstrated that the system could perform autonomous navigation tasks, but accuracy was limited by gait variability, turning imprecision, and sensing constraints.

Localization and planning performance showed measurable positional and orientation error caused primarily by open-loop motion control and mechanical variability. Mapping experiments demonstrated effective environment coverage but revealed inefficiencies due to repeated body rotations for sensing. These findings emphasized the gap between theoretical navigation algorithms and the physical implementations on a real-world robot.

Future improvements should focus on reducing motion error through feedback-based gait control, improving sensing efficiency through rotating sensors, and integrating correction mechanisms that compensate for accumulated drift. Enhancing these aspects would improve navigation accuracy, reduce execution time, and produce more reliable maps. The results of this lab demonstrate both the feasibility and practical challenges of autonomous robot navigation in real-world systems.

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